

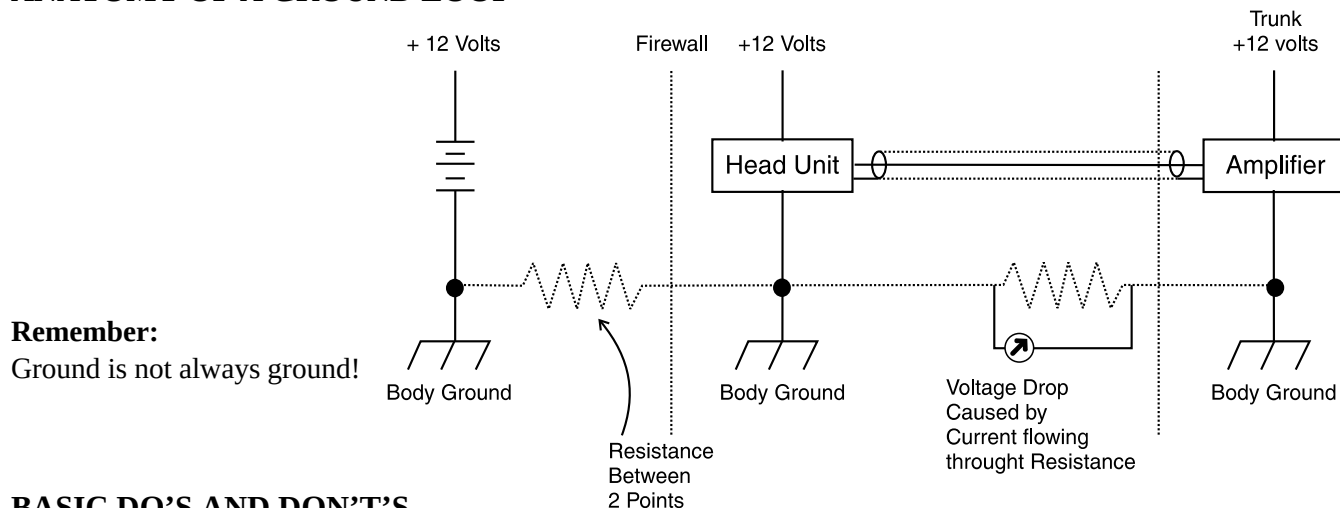
# Tracking Down Ground Loops

## NOISE, WHAT KIND IS IT?

In the world of car stereo, we have to deal with three kinds of unwanted noise—thermal, induced and ground loop. Usually, ground loop noise is engine speed related alternator whine but includes fuel pump, fan and any other electrical components in the car. Induced noise travels through the air from an adjacent wire, power supply or the like. Thermal noise is always present because operating your system at absolute zero temperature is not convenient. In modern day electronics, as evidenced by current signal-to-noise specs, thermal noise is not a problem.

90% of all car stereo noise problems are related to ground loops. Turn this page over for a road map to troubleshooting ground loops.

## ANATOMY OF A GROUND LOOP



## BASIC DO'S AND DON'T'S

You have heard these before, probably, but then again most of us didn't listen to our mother the first time either!

### Do's

1. Use large enough power wire.
2. Run 100% shielded audio cables.
3. Properly level adjust (level match).
4. Diagram the system BEFORE starting ("Pay me now or pay me later")
5. Start with a common star ground point. Consider grounding everything to the negative battery terminal.

### Don't's

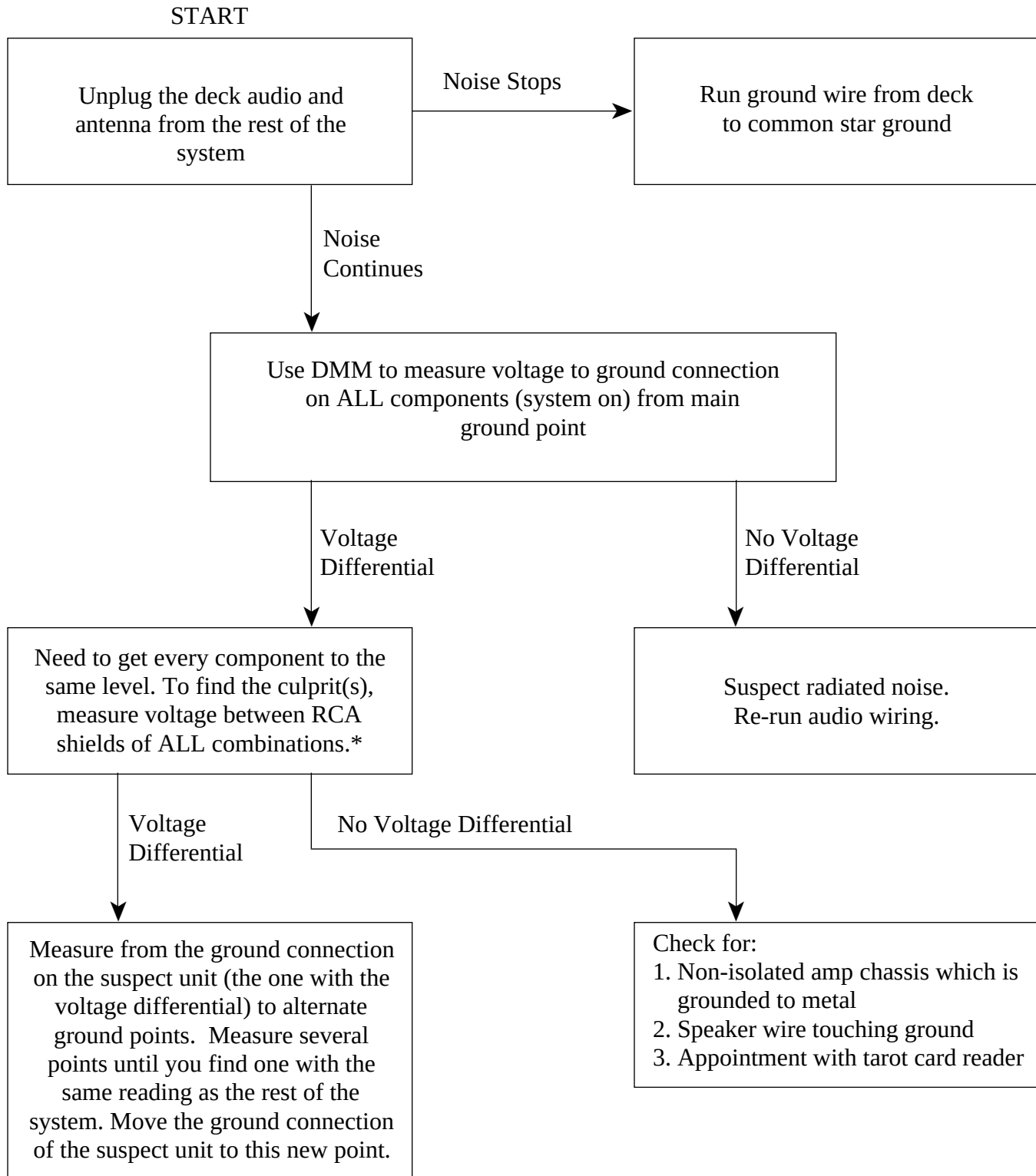
1. Use a ground wire from the OEM wiring harness
2. Run the signal and power wires together
3. Ground the deck only to the dash
4. Use an alternator with bad diodes

## FINAL HINT

The ground loop road map on the back requires using a digital voltmeter (sometimes called a DMM). There are other nifty uses for this tool too numerous to mention (If you don't have one, buy one. Good Fluke meters are less than \$75.00).

One of these nifty uses is to find blown speakers. Besides measuring volts, the DMM measures ohms or impedance. A blown speaker is an open circuit and has infinite impedance. Using the DMM is usually faster (do it back at the amps) than other methods.

## Ground Loop Road Map



\* Do these measurements carefully. Sneaky things can be happening. Often through the signal ground a third unit, is the conduit between two other units that have a ground loop. The third unit may be totally isolated between signal and power (like the Audio Control products). So measure all combinations.